

Bachelor of Technology — Bioengineering

Table 1: Bioengineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	42
	Discipline Electives	24
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Bioengineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	BE-201	Cell Biology	3-0-2-4	4
2	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
3	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
4	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
5	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
6	BE-301	Biomechanics	3-0-2-4	4
7	BE-304	Bioinformatics	3-0-2-4	4
8	BE-101P	Reverse Engineering	0-0-2-1	1
9	BE-303	Applied Biostatistics	3-0-2-4	4
10	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
11	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
12	BE-310	Biomaterials	3-0-2-4	4
Total				42

Table 3: B. Tech Bioengineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Bioengineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Bioengineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	BE-201	Cell Biology	3-0-2-4	4
3	DC	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
4	DC	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
5	DC	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Bioengineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
3	DC	BE-301	Biomechanics	3-0-2-4	4
4	DC	BE-304	Bioinformatics	3-0-2-4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Bioengineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-101P	Reverse Engineering	0-0-2-1	1
2	DC	BE-303	Applied Biostatistics	3-0-2-4	4
3	DC	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
4	DC	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
5	DC	BE-310	Biomaterials	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Bioengineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Bioengineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	BE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Bioengineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	BE-010	Internship		2
2	MTP	BE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 24 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Bioengineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-506	Cognitive Robotics	3
2	BE-101	Reverse Engineering for Bioengineers	3
3	BE-302	Bioelectric Systems Modeling	4
4	BE-307	Reverse Engineering for Bioengineers	3
5	BE-401	Bioengineering Mini Project, Term Paper and Seminar	4
6	BE-501	Anatomy and Physiology	3
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-503	Biosensing & Bioinstrumentation	4
9	BE-504	Biomaterials	4
10	BE-505	Computational Biology	3
11	BE-506	Biological Modelling and Simulation	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	BE-507	Environmental Impact Assessment	3
13	BE-598	Thesis Project Part I	3
14	BE-599	Thesis Project Part II	3
15	BE-698	Thesis Project Part I	3
16	BE-699	Thesis Project Part II	17
17	BY-501	DNA Nanotechnology	3
18	BY-502	Biophysics and Protein Engineering	3
19	BY-503	Cellular Fuel and Cellular Communication	3
20	BY-504	Metabolic Systems Biology	4
21	BY-506	Advanced Immunology	3
22	BY-507	Genetic Engineering	4
23	BY-508	Practical Metabolomics	3
24	BY-509	Practical OMICs	3
25	BY-510	Advanced Cell Biology	3
26	BY-511	Cell Physiology in health and disease	3
27	BY-512	Quantitative and Computational Biology	3
28	BY-514	Analytical Biotechniques	3
29	BY-515	Molecular Biotechnology	3
30	BY-516	Introduction to OMICS and Systems Analysis	3
31	BY-517	Proteomics	3
32	BY-518	Disease Biology	3
33	BY-519	Protein Sciences in Therapeutics	3
34	BY-520	Cell Biology and Physiology Lab	3
35	BY-520P	Cell Biology and Physiology	1
36	BY-521	Computational Biology and Cellular Bioprocess Technology Lab	3
37	BY-521P	Computational Biology and Cellular Bioprocess Technology Lab	1
38	BY-522	Analytical and Molecular Biotechnology Lab	3
39	BY-523	Systems Biology Lab	3
40	BY-524	Medical and NanoBiotechnology Lab	3
41	BY-527	Gene silencing and genome editing: principles and applications	3
42	BY-528	Sensory Biology	3
43	BY-529	Mechanobiology of the Cell (MBoC)	3
44	BY-531	Quantitative Biology and Data Analytics	3
45	BY-532	Immunotechnology	3
46	BY-535P	Analytical Biotechniques Lab	1
47	BY-536P	Immunotechnology Lab	1
48	BY-537	Computational Biology-1	4
49	BY-606	Bioinformatics Applications for Systems	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	BY-613	Metagenomics and Next Generation Sequencing Technologies	3
51	CE-521	Ecology and Environment Microbiology	4
52	CE-559	Biological Wastewater Treatment	3
53	CS-309	Information and Database Systems	4
54	CS-606	Cognitive Modeling	3
55	CS-671	Deep Learning and Applications	4
56	CY-344	Food Chemistry: Processing, Preservation and Storage	3
57	CY-550	Bioinspired Materials	3
58	CY-642	Molecular- and Bio-electronics	3
59	CY-643	Advanced Analytical Techniques	3
60	CY-644	Bioinorganic chemistry	3
61	CY-670	Fluorescence spectroscopy, microscopy and applications	3
62	DS-303	Statistical Foundations of Data Science	3
63	DS-314	Probability Theory and Statistics	4
64	EE-305	Digital Signal Processing	4
65	EE-314	Digital Signal Processing	4
66	EE-516	Biomedical Systems	4
67	EE-574	Biomedical Signal and Image Analysis	3
68	EE-608	Digital Image Processing	4
69	MA-621	Modeling Population Dynamics	3
70	MA-650	Mathematical Models for Infectious Diseases	3
71	ME-303	Heat Transfer	3
72	ME-308	Manufacturing Engineering	3
73	ME-527	Biofluid Dynamics	3
74	ME-612	Introduction to Bio-materials	3
75	ME-622	Biomechanics of Musculoskeletal System	3
76	MT-506	Biomaterials	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	BE-498P	Major Technical Project - I (compulsory; counted as DE)	4
78	BE-499P	Major Technical Project - II (compulsory; counted as DE)	4